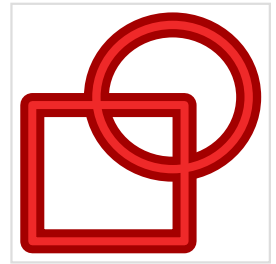


Sketcher Workbench

[← !\[\]\(666e09182d4cd268646ea700ea60dcdf_img.jpg\) Robot Workbench](#)[Index !\[\]\(c3d993ca47bfe2a953c700506ce31fa0_img.jpg\)](#)[☰ Spreadsheet Workbench →](#)

Introduction

With the  [Sketcher Workbench](#) 2D sketches intended for use in other workbenches can be created. 2D sketches are the starting point for many CAD models. They typically define the profiles and paths for operations to create 3D shapes. A model may depend on several sketches for its final shape.

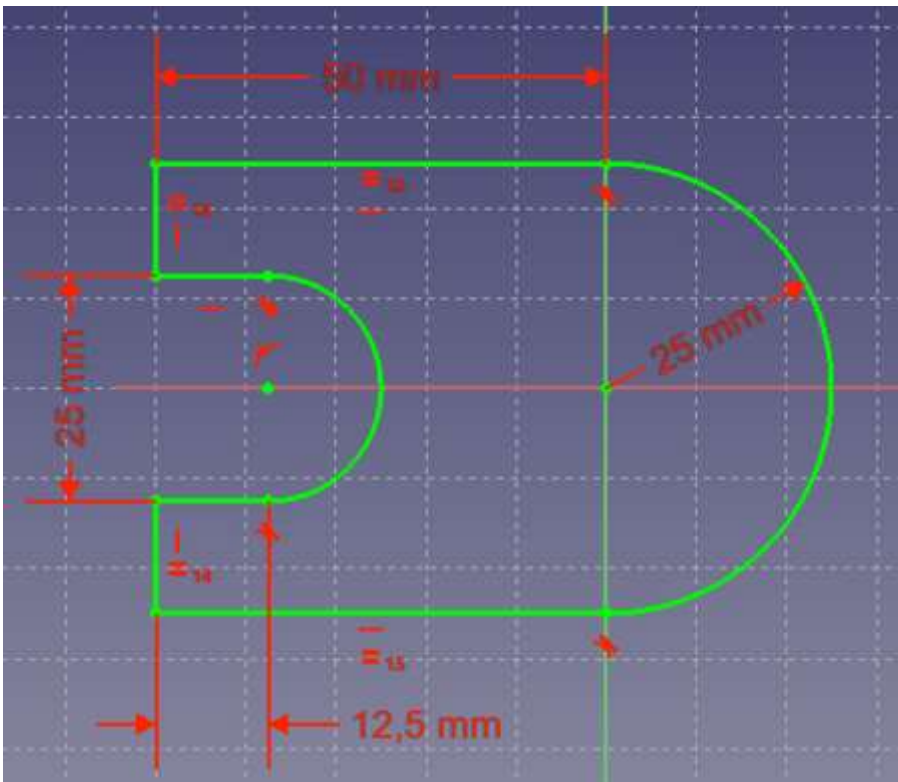


Sketcher workbench
icon

Together with boolean operations defined in the  [Part Workbench](#), the Sketcher Workbench, or "The Sketcher" for short, forms the basis of the [constructive solid geometry \(CSG\)](#) method of building solids. Together with  [PartDesign Workbench](#) operations, it also forms the basis of the [feature editing methodology](#) of creating solids. But many other workbenches use sketches as well.

The Sketcher workbench features [constraints](#), allowing 2D shapes to follow precise geometrical definitions in terms of length, angles, and relationships (horizontality, verticality, perpendicularity, etc.). A constraint solver calculates the constrained-extent of 2D geometry and allows interactive exploration of the degrees-of-freedom of the sketch.

The Sketcher is not intended for producing 2D blueprints. Once sketches are used to generate a solid feature, they are automatically hidden and constraints are only visible in Sketch edit mode. If you only need to produce 2D views for print, and don't want to create 3D models, check out the [Draft workbench](#).



A fully constrained sketch

Constraints

Constraints are used to limit the degrees of freedom of an object. For example, a line without constraints has 4 degrees of freedom (abbreviated as "DoF"): it can be moved horizontally or vertically, it can be stretched, and it can be rotated.

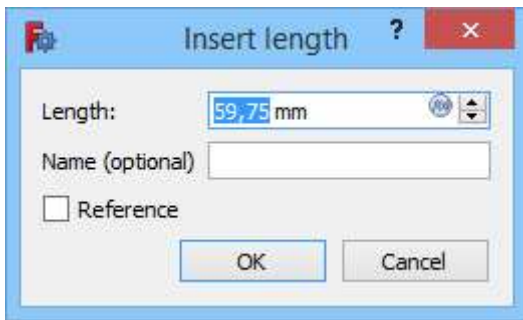
Applying a horizontal or vertical constraint, or an angle constraint (relative to another line or to one of the axes), will limit its capacity to rotate, thus leaving it with 3 degrees of freedom. Locking one of its points in relation to the origin will remove another 2 degrees of freedom. And applying a distance dimension will remove the last degree of freedom. The line is then considered **fully constrained**.

Objects can be constrained in relation to one another. Two lines can be joined through one of their points with a coincident constraint. An angle can be set between them, or they can be set perpendicular. A line can be tangent to an arc or a circle, and so on. A Sketch may have a number of different solutions, and making it **fully constrained** can mean that just one of these possible solutions has been reached based on the applied constraints. See also: Flipping.

There are two kinds of constraints: geometric and dimensional. They are detailed in the Tools section below.

Edit constraints

When a driving dimensional constraint is created, and if the **Ask for value after creating a dimensional constraint** preference is selected (default), a dialog opens to edit its value.



You can enter a numerical value or an expression, and it is possible to name the constraint to facilitate its use in other expressions. You can also check the **Reference** checkbox to switch the constrain to reference mode.

To edit the value of an existing dimensional constraint do one of the following:

- Double-click the constraint value in the 3D View.
- Double-click the constraint in the Sketcher Dialog.
- Right-click the constraint in the Sketcher Dialog and select the **Edit Value** option from the context menu.

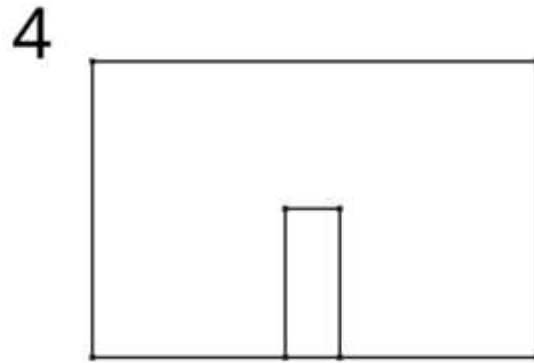
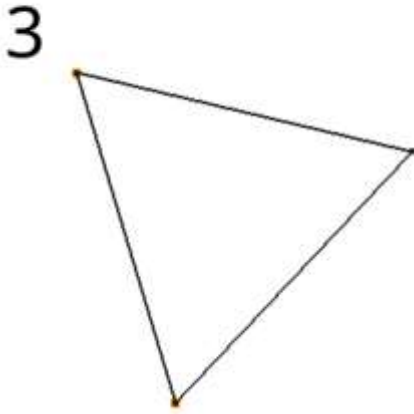
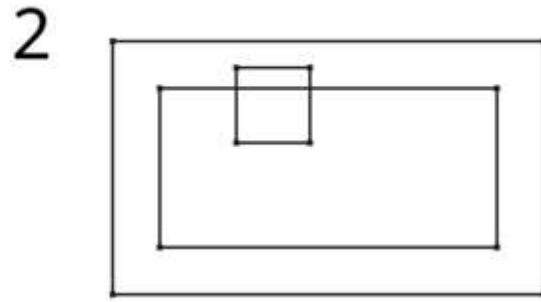
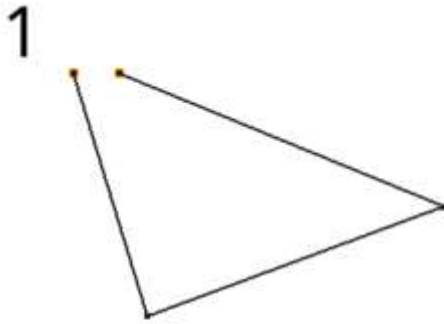
Reposition constraints

Dimensional constraints can be repositioned in the 3D View by dragging. Hold down the left mouse button over the constraint value and move the mouse. The symbols of geometric constraints are positioned automatically and cannot be moved.

Profile sketches

To create a sketch that can be used as a profile for generating solids certain rules must be followed:

1. The sketch must contain only closed contours. Gaps between endpoints, however small, are not allowed.
2. Contours can be nested, to create voids, but should not self-intersect or intersect other contours.
3. Contours cannot share edges with other contours. Duplicate or overlapping edges are not allowed.
4. T-connections, that is more than two edges sharing a common point, or a point touching an edge, are not allowed.



Invalid sketches:

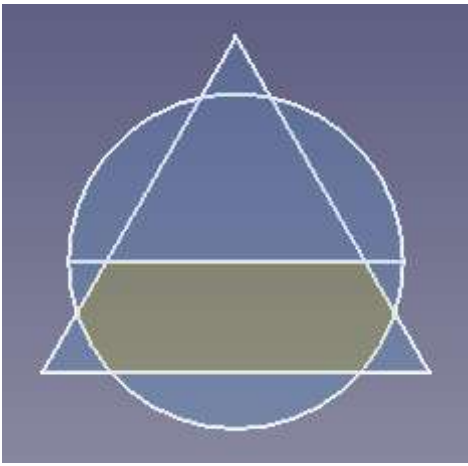
1. Open contour (free endpoints highlighted by the Validate Sketch tool)
2. Intersecting contours
3. Duplicate edges (endpoints of overlapping edges highlighted by the Validate Sketch tool)
4. T-connections

These rules do not apply to construction geometry (default color blue), which is not shown outside edit mode, or if the sketch is used for a different purpose. Depending on the workbench and the tool that will use the profile sketch, additional restrictions may apply.

Internal faces

introduced in 1.0

If the Data **Make Internals** property of a sketch is set to true internal faces are generated based on the edges in the sketch. These faces can be selected as closed profiles for operations. Intersecting contours and T-connections are allowed, they just result in additional faces.



Tools

The Sketcher Workbench tools are located in the Sketch menu and/or several toolbars. Almost all Sketcher toolbars are only displayed while a sketch is in edit mode. The only exception is the Sketcher toolbar which is only displayed if no sketch is in edit mode.

Some tools are also available from the 3D View context menu while a sketch is in edit mode, or from the context menus of the Sketcher Dialog.

If a sketch is in edit mode the Structure toolbar is hidden as none of its tools can then be used.

General

Sketcher toolbar

-  New Sketch: Creates a new sketch and opens the Sketcher Dialog to edit it.
-  Edit Sketch: Opens the Sketcher Dialog to edit an existing sketch.
-  Attach Sketch: Attaches a sketch to selected geometry.
-  Reorient Sketch: Places a sketch on one of the main planes with an optional offset. It can also be used to detach a sketch.
-  Validate Sketch: Can analyze and repair a sketch that is no longer editable or has invalid constraints, or add missing coincident constraints.
-  Merge Sketches: Merges two or more sketches.

-  Mirror Sketch: Mirrors sketches across their X-axis, Y-axis, or origin.

Edit Mode toolbar

-  Leave Sketch: Finishes sketch edit mode and closes the Sketcher Dialog.
-  Cancel Editing: Leaves sketch edit mode, reverts any changes, and closes the Sketcher Dialog. introduced in 26.3
-  Align View to Sketch: Aligns the 3D View with the sketch.
-  Toggle Section View: Toggles a temporary section plane that hides any objects and parts of objects in front of the sketch plane.

Other

-  Stop Operation: Stops any currently running geometry or constraint creation tool.
- Grid: Grid settings can be changed in the menu of the Sketch Edit section of the Sketcher Dialog.
- Snap: Snap settings can be changed in the same menu.
- Rendering Order: The rendering order can be changed in the same menu.

Geometries

These are tools for creating objects.

-  Point: Creates a point.
-  Polyline: Creates a series of line and arc segments connected by their endpoints. The tool has several modes.
-  Line: Creates a line. introduced in 1.0: The tool has three modes.
-  Create Arc:
 -  Arc From Center: Creates an arc from its center and its endpoints. introduced in 1.0: Or from its endpoints and a point along the arc.

-  Arc From 3 Points: Creates an arc from its endpoints and a point along the arc. introduced in 1.0: This is the same tool as Arc From Center but with a different initial mode.
-  Elliptical Arc: Creates an elliptical arc.
-  Hyperbolic Arc: Creates an hyperbolic arc.
-  Parabolic Arc: Creates an parabolic arc.
-  Create Circle/Ellipse:
 -  Circle From Center: Creates a circle from its center and a point along the circle. introduced in 1.0: Or from three points along the circle.
 -  Circle From 3 Points: Creates a circle from three points along the circle. introduced in 1.0: This is the same tool as Circle From Center but with a different initial mode.
 -  Ellipse From Center: Creates an ellipse from its center, an endpoint of one of its axes, and a point along the ellipse. introduced in 1.0: Or from both endpoints of one of its axes and a point along the ellipse.
 -  Ellipse From 3 Points: Creates an ellipse from the endpoints of one of its axes and a point along the ellipse. introduced in 1.0: This is the same tool as Ellipse From Center but with a different initial mode.
-  Create Rectangle:
 -  Rectangle: Creates a rectangle. introduced in 1.0: The tool has four modes. Rounded corners and creating an offset copy are optional features.
 -  Centered Rectangle: Creates a centered rectangle. introduced in 1.0: This is the same tool as Rectangle but with a different initial mode.
 -  Rounded Rectangle: Creates a rounded rectangle. Idem.
-  Create Polygon:

-  Triangle: Creates an equilateral triangle. introduced in 1.0: This is the same tool as Polygon but with the number of sides preset to a specific value.
-  Square: Creates a square. Idem.
-  Pentagon: Creates a pentagon. Idem.
-  Hexagon: Creates a hexagon. Idem.
-  Heptagon: Creates a heptagon. Idem.
-  Octagon: Creates an octagon. Idem.
-  Polygon: Creates a regular polygon. The number of sides can be specified.
-  Create Slot:
 -  Slot: Creates a slot.
 -  Arc Slot: Creates an arc slot. introduced in 1.0
-  Create B-Spline:
 -  B-Spline: Creates a B-spline curve from control points. introduced in 1.0: Or from knot points.
 -  Periodic B-Spline: Creates a periodic (closed) B-spline curve from control points. introduced in 1.0: This is the same tool as B-Spline but with a different initial mode.
 -  B-Spline From Knots: Creates a B-spline curve from knot points. Idem.
 -  Periodic B-Spline From Knots: Creates a periodic (closed) B-spline curve from knot points. Idem.

-  Text: Creates text geometry controlled by a text constraint. This is an experimental feature that may change in future versions. [introduced in 26.3](#)
-  Toggle Construction Geometry: Either toggles the geometry creation tools to/from construction mode, or toggles selected geometry to/from construction geometry.

Constraints

These are tools for creating constraints. Some constraints require the use of helper constraints.

-  Dimensional Constraints:
 -  Dimension: Is the context-sensitive constraint tool of the Sketcher Workbench. Based on the current selection, it offers appropriate dimensional constraints, but also geometric constraints. [introduced in 1.0](#)
 -  Horizontal Dimension: Fixes the horizontal distance between two points or the endpoints of a line. If a single point is pre-selected, the distance is relative to the origin of the sketch.
 -  Vertical Dimension: Fixes the vertical distance between two points or the endpoints of a line. If a single point is pre-selected, the distance is relative to the origin of the sketch.
 -  Distance Dimension: Fixes the length of a line, the distance between two points, the perpendicular distance between a point and a line; or the distance between the edges of two circles or arcs, or between the edge of a circle or arc and a line; or, [introduced in 1.0](#), the length of an arc.
 -  Radius/Diameter Dimension: Fixes the radius of arcs and B-spline weight circles, and the diameter of circles.
 -  Radius Dimension: Fixes the radius of circles, arcs and B-spline weight circles.
 -  Diameter Dimension: Fixes the diameter of circles and arcs.

-  Angle Dimension: Fixes the angle between two edges, the angle of a line with the horizontal axis of the sketch, or the aperture angle of a circular arc.
-  Lock Position: Applies horizontal dimension and vertical dimension constraints to points. If a single point is selected the constraints reference the origin of the sketch. If two or more points are selected the constraints reference the last point in the selection.
-  Coincident Constraint (Unified): Creates a coincident constraint between points, fixes points on edges or axes, or creates a concentric constraint. It combines the Coincident Constraint and Point-On-Object Constraint tools. introduced in 1.0
-  Coincident Constraint: Creates a coincident constraint between points, or a concentric constraint.
-  Point-On-Object Constraint: Fixes points on edges or axes.
-  Horizontal/Vertical Constraints:
 -  Horizontal/Vertical Constraint: Constrains lines or pairs of points to be horizontal or vertical, whichever is closest to the current alignment. It combines the Horizontal Constraint and Vertical Constraint tools. introduced in 1.0
 -  Horizontal Constraint: Constrains lines or pairs of points to be horizontal.
 -  Vertical Constraint: Constrains lines or pairs of points to be vertical.
-  Parallel Constraint: Constrains lines to be parallel.
-  Perpendicular Constraint: Constrains two lines to be perpendicular, or two edges, or an edge and an axis, to be perpendicular at their intersection. The constraint can also connect two edges, forcing them to be perpendicular at the joint.
-  Tangent/Collinear Constraint: Constrains two edges, or an edge and an axis, to be tangent. The constraint can also connect two edges, forcing

them to be tangent at the joint. If two lines are selected they are made collinear.

-  Equal Constraint: Constrains edges to have an equal length (lines) or curvature (other edges except B-splines).
-  Symmetric Constraint: Constrains two points to be symmetrical around a line or axis, or around a third point.
-  Block Constraint: Blocks edges in place with a single constraint. It is mainly intended for B-splines.
-  Group Constraint: Groups selected geometry. This is an experimental feature that is not enabled by default. [introduced in 26.3](#)
-  Refraction Constraint: Constrains two lines to follow the law of refraction of light as it penetrates through an interface.
-  Toggle Constraints:
 -  Toggle Driving/Reference Constraints: Toggles the dimensional constraint creation tools between driving and reference mode, or toggles selected dimensional constraints between those modes.
 -  Toggle Constraints: Activates or deactivates selected constraints.

Sketcher Tools

-  Create Fillet/Chamfer:
 -  Fillet: Creates a fillet between two non-parallel edges. [introduced in 1.0](#):
The tool can also create a chamfer.
 -  Chamfer: creates a chamfer between two non-parallel edges. This is the same tool as Fillet but with a different initial mode. [introduced in 1.0](#)
-  Edit Edge:
 -  Trim Edge: Trims an edge at the nearest intersections with other edges.
 -  Split Edge: Splits an edge while transferring most constraints.

-  Extend Edge: Extends or shortens a line or an arc to an arbitrary location, or to a target edge or point.
-  External Geometry:
 -  External Projection: Creates the projection edges of external geometry. introduced in 1.1
 -  External Intersection: Creates the intersection edges of external geometry with the sketch plane. introduced in 1.1
-  Carbon Copy: Copies all geometry and constraints from another sketch into the active sketch.
-  Select Origin: Selects the origin of the sketch.
-  Select Horizontal Axis: Selects the horizontal axis of the sketch.
-  Select Vertical Axis: Selects the vertical axis of the sketch.
-  Move/Array Transform: Moves or optionally creates copies of selected elements. introduced in 1.0
-  Rotate/Polar Transform: Rotates or optionally creates rotated copies of selected elements. introduced in 1.0
-  Scale: Scales or optionally creates scaled copies of selected elements. introduced in 1.0
-  Offset: Creates equidistant edges around selected edges. introduced in 1.0
-  Mirror: Creates mirrored copies of selected elements.
-  Remove Axes Alignment: Removes the axes alignment of selected edges by replacing horizontal and vertical constraints with parallel and perpendicular constraints.
-  Delete All Geometry: Deletes all geometry and all constraints from the sketch.
-  Delete All Constraints: Deletes all constraints from the sketch.
-  Copy Elements: See Copy, cut and paste.

-  Cut Elements: See Copy, cut and paste.
-  Paste Elements: See Copy, cut and paste.

B-Spline Tools

-  Geometry to B-Spline: Converts edges to B-splines.
-  Increase B-Spline Degree: Increases the degree (order) of B-splines.
-  Decrease B-Spline Degree: Decreases the degree (order) of B-splines.
-  Increase Knot Multiplicity: Increases the multiplicity of a B-spline knot.
-  Decrease Knot Multiplicity: Decreases the multiplicity of a B-spline knot.
-  Insert Knot: Inserts a knot into a B-spline or increases the multiplicity of an existing knot.
-  Join Curves: Creates a B-spline by joining two existing B-splines or other edges.

Visual Helpers

-  Select Under-Constrained Elements: Selects the not fully constrained elements in the sketch.
-  Select Associated Constraints: Selects the constraints associated with sketch elements.
-  Select Associated Geometry: Selects the sketch elements associated with constraints.
-  Select Redundant Constraints: Selects the redundant constraints in the sketch.
-  Select Conflicting Constraints: Selects the conflicting constraints in the sketch.
-  Toggle Circular Helper for Arcs: Shows or hides the circular helpers (underlying virtual circles) for arcs in all sketches. introduced in 1.0
-  Toggle B-Spline Information Layer:

-  Toggle B-Spline Degree: Shows or hides the B-spline degree in all sketches.
-  Toggle B-Spline Control Polygon: Shows or hides the B-spline control polygon in all sketches.
-  Toggle B-Spline Curvature Comb: Shows or hides the B-spline curvature comb in all sketches.
-  Toggle B-Spline Knot Multiplicity: Shows or hides the B-spline knot multiplicity in all sketches.
-  Toggle B-Spline Control Point Weight: Shows or hides the B-spline control point weight in all sketches.
-  Toggle Internal Geometry: Deletes the internal geometry of elements, or recreates missing internal geometry.
-  Switch Virtual Space: (un)hides constraints or switches the visible virtual space.

Obsolete tools

-  Clone: Clones a Sketcher element. Not available in the default GUI in 1.0 and above. Can still be added to a custom toolbar.
-  Close shape: Creates a closed shape by applying coincident constraints to endpoints. Not available in 0.21 and above.
-  Corner-preserving fillet: Creates a fillet between two non-parallel lines while preserving their corner point. Not available in 1.0 and above.
-  Connect edges: Connect Sketcher elements by applying coincident constraints to endpoints. Not available in 0.21 and above.
-  Copy: Copies a Sketcher element. Not available in the default GUI in 1.0 and above. Can still be added to a custom toolbar.
-  External geometry: Projects edges and/or vertices belonging to objects outside the sketch onto the sketch plane. Not available in 1.1 and above.

-  Move: Moves the selected geometry taking as reference the last selected point. Not available in the default GUI in 1.0 and above. Can still be added to a custom toolbar.
-  Polyline (legacy tool): Creates a series of line and arc segments connected by their endpoints. Not available in the default GUI in 26.3 and above. Can still be added to a custom toolbar.
-  Rectangular array: Creates an array of selected Sketcher elements. Not available in the default GUI in 1.0 and above. Can still be added to a custom toolbar.

Preferences

-  Preferences: Preferences for the Sketcher Workbench.

Drawing aids

The Sketcher Workbench has several drawing aids and other features that can help when creating geometry and applying constraints.

Continue modes

There are two continue modes: **Geometry creation "Continue Mode"** and **Constraint creation "Continue Mode"**. If these are checked (default) in the preferences, related tools will restart after finishing. To exit a continuous tool press Esc or the right mouse button. This must be repeated if a continuous geometry tool has already received input. You can also exit a continuous tool by starting another geometry or constraint creation tool. Note that pressing Esc if no tool is active will exit sketch edit mode. Uncheck the **Esc key can leave sketch edit mode** preference if you often inadvertently press Esc too many times.

Auto constraints

In sketches that have **Auto constraints** checked (default) several constraints are applied automatically when creating or (introduced in 26.3) dragging geometry. The icon of a proposed automatic constraint is shown next to the cursor when it is placed correctly. Left-clicking, or ending the drag operation by releasing the Left mouse button, will then apply that constraint. This is a per-sketch setting that can be changed in the Sketcher Dialog or by changing the Autoconstraints property of the sketch.

The following constraints are applied automatically:

-  Coincident

-  Point-on-object
-  Horizontal
-  Vertical
- introduced in 26.3:  Parallel (see Directional helpers)
- introduced in 26.3:  Perpendicular (idem)
-  Tangent
- introduced in 1.0:  Symmetric (line midpoint)

Circular helpers

To create auto constraints along extended arcs circular helpers can be shown.

Linear helpers

introduced in 26.3

Lines are automatically extended with helper lines if the cursor is positioned correctly. This allows creating  point-on-object auto constraints along the line.

Directional helpers

introduced in 26.3



Using a directional helper line to create a perpendicular auto constraint

If the **Show directional autoconstraint helpers** preference is selected (default), tangential, or perpendicular and parallel helper lines referencing a target edge can be shown while using the  Polyline tool (only when creating line segments) and  Line tool.

Note that straight target edges that are horizontal or vertical are ignored.

1. Pick the first point of a new line (segment).
2. If this point is auto constrained to be  coincident with the endpoint of an existing line or circular arc, that edge will be the target edge. A previous segment of a polyline automatically

becomes the target edge as well.

3. If there is no target edge yet: hover an existing straight edge.
4. A tangential helper line, or a perpendicular and a parallel helper line appear around the first point.
5. Optionally change the target edge by hovering a different straight edge. This will realign the helper lines.
6. Snap to a helper line to create a  tangent,  parallel or  perpendicular constraint between the new line and the target edge.
7. For a perpendicular constraint you can also pick a point on the target edge itself. In that case a  point-on-object or a  coincident constraint will be created as well.

Snapping

It is possible to snap to grid lines and grid intersection, and to snap to edges of geometry and midpoints of lines and arcs, and to certain angles. Please note that snapping does not produce constraints in and of itself. For example, only if Auto constraints is switched on will snapping to an edge produce a point-on-object constraint. But just picking a point on the edge would then have the same result.

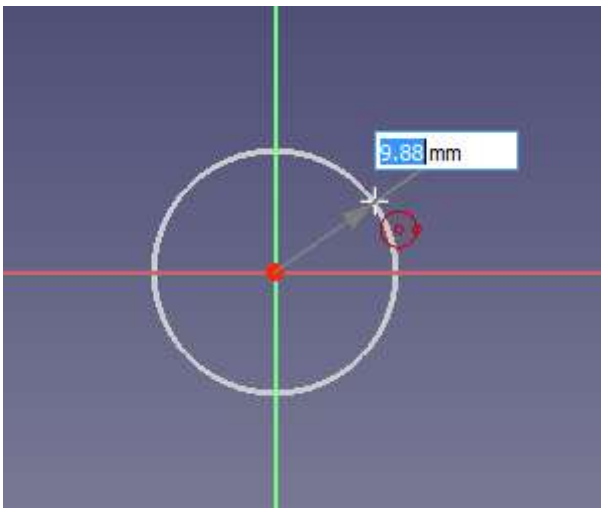
On-View-Parameters

introduced in 1.0

Depending on the selected option in the preferences only the dimensional On-View-Parameters or both the dimensional and the positional On-View-Parameters can be enabled. Positional parameters allow the input of exact coordinates, for example the center of a circle, or the start point of a line. Dimensional parameters allow the input of exact dimensions, for example the radius of a circle, or the length and angle of a line. On-View-Parameters are not available for all tools.



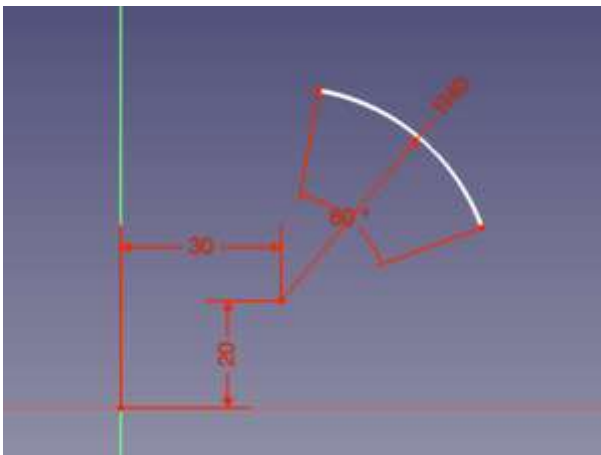
Determining the center point of a circle with the positional parameters enabled



Determining the radius of a circle with the dimensional parameters enabled

If values are entered and confirmed by pressing **Enter** or **Tab**, related constraints are added automatically. If two parameters are displayed at the same time, for example the X- and Y-coordinate of a point, it is possible to enter one value and pick a point to define the other. Depending on the object additional constraints may be required to fully constrain it. Constraints resulting from On-View-Parameters take precedence over those that may result from Auto constraints.

introduced in 26.3 If an OVP input has the focus you can hold down **Alt** to send key strokes to the 3D View instead.



Arc created by entering all On-View-Parameters with resulting automatically created constraints

Coordinate display

If the **Show coordinates next to the cursor while editing** preference is checked (default), the parameters of the current geometry tool (coordinates, radius, or length and angle) are displayed next to the cursor. This is deactivated while On-View-Parameters are shown.

Selection methods

While a sketch is in edit mode several selection methods can be used. introduced in 26.3: For all methods selection filters are taken into account.

3D View element selection

As elsewhere in FreeCAD, an element can be selected in the 3D View with a single left mouse click. But there is no need to hold down the **Ctrl** key when selecting multiple elements. Holding down that key is possible though and has the advantage that you can miss-click without losing the selection. Edges, points and constraints can be selected in this manner.

3D View box selection

Box selection in the 3D View works without using Std BoxSelection or Std BoxElementSelection:

1. Make sure that no tool is active.
2. Do one of the following:
 - Click in an empty area and drag a rectangle from left to right to select elements that lie completely inside the rectangle.
 - Click in an empty area and drag a rectangle from right to left to also select elements that touch or cross the rectangle.

You can box-select edges and points, constraints cannot be box-selected.

3D View connected geometry selection

introduced in 1.0

Double-clicking an edge in the 3D View will select all edges directly and indirectly connected with that edge via endpoints. There is no need for the edges to be connected with coincident constraints, endpoints need only have the same coordinates.

Sketcher Dialog selection

Edges and points can also be selected from the Elements section of the Sketcher Dialog, and constraints from the Constraints section of that dialog.

Select All shortcut

To select everything within a sketch, use the standard keyboard shortcut **Ctrl**+**A** or the **Edit** → **Select All** option from the menu.

Copy, cut and paste

introduced in 1.0

The standard keyboard shortcuts, **Ctrl**+**C**, **Ctrl**+**X** and **Ctrl**+**V**, can be used to copy, cut and paste selected Sketcher geometry including related constraints. But these tools are also available from the **Sketch** → **Sketcher Tools** menu. They can be used within the same sketch but also between different sketches or separate instances of FreeCAD. Since the data is copied to the clipboard in the form of Python code, it can be used in other ways too (e.g. shared on the forum).

Best practices

Every CAD user develops their own way of working over time, but there are some useful general principles to follow.

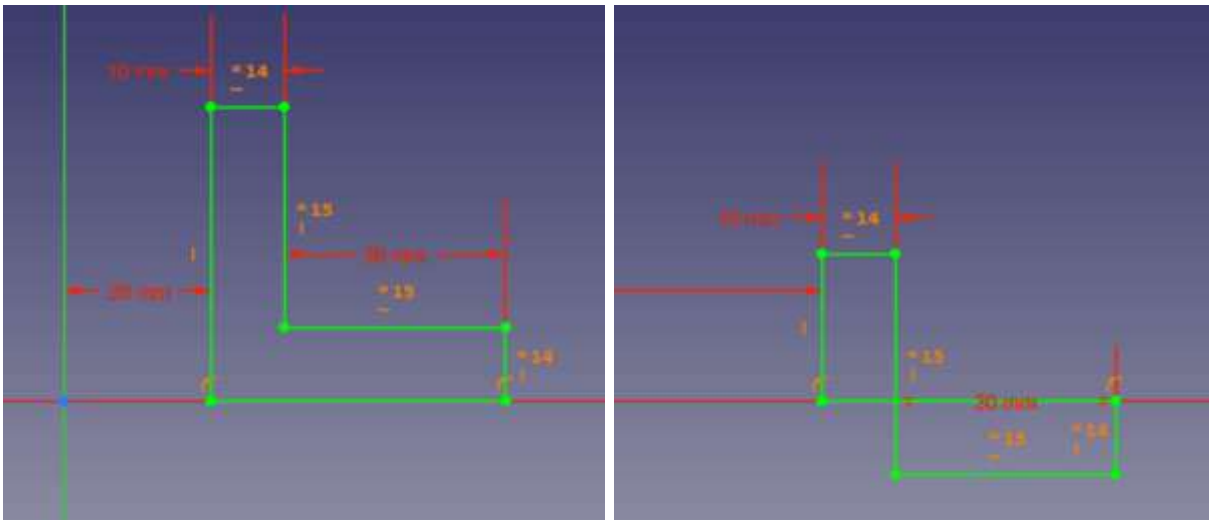
- A series of simple sketches is easier to manage than a single complex one. For example, a first sketch can be created for the base 3D feature (a pad or a revolve for example), while a second one can contain holes or cutouts (pockets). Some details can be left out, to be realized later on as 3D features. You can choose to avoid fillets in your sketch if there are too many, and add them as a 3D feature.
- Always create a closed profile, or your sketch won't produce a solid, but rather a set of open faces. If you don't want some of the objects to be included in the solid creation, turn them to construction elements with the [Toggle Construction Geometry](#) tool.
- Use the Auto constraints feature to limit the number of constraints you'll have to add manually.
- As a general rule, apply geometric constraints first, then dimensional constraints, and lock your sketch last. But remember: rules are made to be broken. If you're having trouble manipulating your sketch, it may be useful to constrain a few objects first before completing your profile.
- If possible, center your sketch at the origin (0,0). If your sketch is not symmetric, locate one of its points at the origin.
- If you have the possibility to choose between the distance dimension and the horizontal or vertical dimensions, prefer the latter. Horizontal and vertical dimensions are computationally cheaper.
- In general, the best constraints to use are: horizontal and vertical constraints; horizontal and vertical dimensions; point-to-point tangent

constraints. If possible, limit the use of these: distance dimension; edge-to-edge tangent constraint; point-on-object constraint; symmetric constraint.

- If in doubt about the validity of a sketch once it is complete (features turn green), close the Sketcher Dialog and use  Validate Sketch.

Flipping

The phenomenon that a fully constrained sketch, usually after a major change to one of its dimensions, reaches an unintended new state is known as "flipping". In the example below, changing one dimension completely changes the shape of the sketch. Note that the sketch with the new shape is still fully constrained.



Original sketch (left), and the same sketch after increasing the 20mm value to 1000mm (right)

Methods to reduce flipping

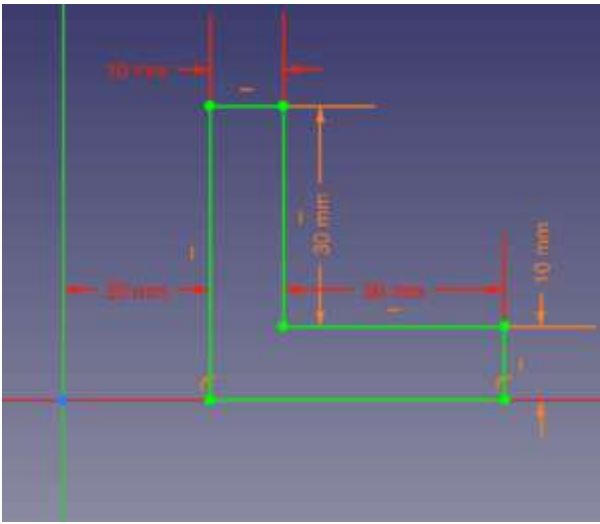
Change dimensions in small increments

This is not always practical, but changing dimension values in smaller increments can work.

Use a different solver

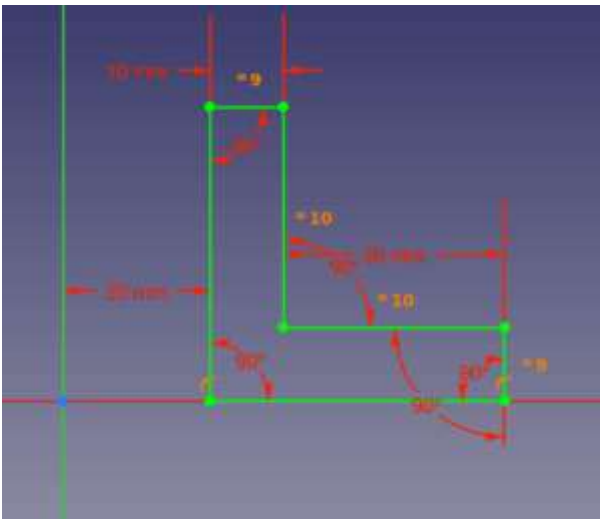
The LevenbergMarquardt solver, which is not the default solver, is known to be less prone to flipping. See Sketcher Dialog for more information.

Use horizontal and vertical dimensions



Using horizontal and vertical dimensions instead of distance dimensions or equal constraints can prevent flipping. Points constrained by these dimensions will only switch places if the sign of their value is changed. In the image above the added (orange) dimensional constraints are linked to the original dimensions via expressions.

Use angle dimensions



Using angle dimensions instead of horizontal and vertical constraints can also work. The angle between edges constrained by angle dimensions will not change. 180° will not become 0° , 90° will not become 270° , etc. In the image all horizontal and vertical constraints have been replaced, but just replacing two is already effective here.

Tutorials

- Sketcher Lecture (<https://forum.freecad.org/viewtopic.php?f=36&t=30104>) by chrisb. This is a more than 80 page PDF document that serves as a detailed manual for the Sketcher. It explains the basics of Sketcher usage, and goes into a lot of detail about the creation of geometrical shapes, and each of the constraints.
- Basic Sketcher Tutorial for beginners

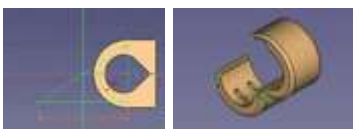
- [Sketcher Micro Tutorial - Constraint Practices](#)
- [Sketcher requirement for a sketch](#) Minimum requirement for a sketch and Complete determination of a sketch

Scripting

The [Sketcher scripting](#) page contains examples on how to create constraints from Python scripts.

Examples

For some ideas of what can be achieved with Sketcher tools, have a look at: [Sketcher Examples](#).



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Estratto da "http://wiki.freecad.org/index.php?title=Sketcher_Workbench&oldid=1856464"